

Original Article

Elucidation of molecular mechanism for colistin resistance among Gram-negative isolates from tertiary care hospitals

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ABSTRACT

Antimicrobial resistance is a growing concern of global public health. The emergence of colistin-resistance among carbapenem-resistant (CPR) Gram-negative bacteria causing fear of pan-resistance, treatment failure, and high mortality across the globe.

Aim: To determine the genotypic colistin-resistance mechanisms among colistin-resistant (CR) Gram-negative clinical isolates along with genomic insight into hypermucoviscous (*hv*)-CR-*Klebsiella pneumoniae*.

Methods: Phenotypic colistin-resistance via broth-microdilution method. PCR-based detection of plasmid-mediated colistin resistance genes (*mcr*-1,2,3). Characterization of selected *hv*CR-*K. pneumoniae* via Whole-genome sequencing.

Results: Phenotypic colistin-resistance was 28% among CPR-Gram-negative isolates of which 90% of CR-isolates displayed MDR profile with overall low plasmid-mediated colistin resistance (*mcr*-2 = 9.4%; *mcr*-3 = 6%). Although *K. pneumoniae* isolates showed the highest phenotypic colistin-resistance (51%) however, relatively low plasmid-mediated gene-carriage (*mcr*-2 = 11.5%; *mcr*-3 = 3.4%) pointed toward other mechanisms of colistin-resistance. *mcr*-negative CR-*K. pneumoniae* displaying *hv*-phenotype were subjected to WGS. *In-silico* analysis detected 7-novel mutations in lipid-A modification genes includes *eptA*(I38V; V50L; A135P), *opgE*(M53L; T486A; G236S), and *arnD*(S164P) in addition to several non-synonymous mutations in lipid-A modification genes conferring resistance to colistin. Insertion of 6.6-kb region harboring putative-PEA-encoding gene (*yjgX*) was detected for the first time in *K. pneumoniae* (*hv*CRKP4771). *In-silico* analysis further confirmed the acquisition of not only MDR determinants but several hypervirulent-determinants displaying a convergent phenotype.

Conclusion: overall high prevalence of phenotypic colistin resistance but low *mcr*-gene carriage suggested complex chromosomal mediated resistance mechanism especially in *K. pneumoniae* isolates. The presence of novel mutations in lipid-A modification genes and the acquisition of putative-PEA-encoding gene by *hv*CR-*K. pneumoniae* points toward the role of chromosomal determinants conferring resistance to colistin in the absence of *mcr*-genes.

1. Introduction

Antimicrobial resistance (AMR) is a growing concern of public health globally, as infections caused by Multidrug-resistant (MDR) Gram-negative bacteria restrict treatment choices and has led to the reuse of ancient drug-like Polymyxins [1]. After the development of carbapenem resistance by MDR Gram-negative bacteria revival of Colistin

(polymyxin-E) was considered as the last drug in the available antibiotic armament for the treatment of CPR-Gram-negative infections [2]. Despite being potent against the MDR pathogens, the non-judicious use of colistin in agriculture, poultry, veterinary fields, and even in clinical settings has resulted in the development of colistin resistance among pathogens [3]. This widespread use of colistin also contributed to resistance in CPR Gram-negative pathogens via diverse chromosomal

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and plasmid-mediated mechanisms which are now considered to be the cause of pan-resistance, treatment failure, and high death rate worldwide [4]. Historically, colistin resistance was considered mainly due to chromosomal acquisition of mutations either in two-component system genes or genes that modify lipopolysaccharide (LPS) by the addition of phospho-ethanol-amine (PEA) or 4-amino-4-deoxy-L-arabinose (Lipid A-Ara4N) to lipid-A, reducing the binding of colistin [5]. But in 2016, ‘*mcr-1*’ (mobile colistin resistance gene), a plasmid-borne gene that encodes PEA-transferase was discovered in China [6]. Several minor variants of *mcr-1* gene also emerged along with other transferable colistin-resistant genes like *mcr-2* to *mcr-9* with a more recent addition of *mcr-10* [7]. Sporadic studies reported a higher prevalence of colistin resistance from Southeast Asia (13.8%–50%) as compared to Europe (10%) [8]. However, data related to colistin resistance is scarce from Southeast Asia as there are limited reports on surveillance of antimicrobial resistance in health care settings.

Pakistan is among Southeast Asian countries tackling the burden of infections due to MDR Gram-negative bacteria which is becoming a grave health concern [9,10]. The antimicrobial data from several studies highlighted the development of colistin resistance among Gram-negative pathogens but the recent upsurge of colistin resistance raised serious concern for the wide-scale dissemination of colistin resistance in our hospital settings [12,13,16]. Colistin resistance in Pakistan is reported to be mediated via plasmid-mediated colistin resistance *mcr* genes, however, its carriage is low despite high phenotypic colistin-resistant Gram-negative bacteria in human clinical samples [16–18]. This points toward other mechanisms of colistin resistance that are neglected so far and need to be explored in our clinical settings. This necessitates a need for conducting a comprehensive study to elucidate both chromosomal and plasmid mediated mechanisms of colistin resistance among phenotypic CR isolates from local hospital settings.

2. Material and methods

2.1. Selection criteria

As colistin is the last drug of choice for the treatment of CPR bacteria therefore only those isolates that were resistant to three of the carbapenem drugs namely imipenem, meropenem and ertapenem were selected for the study. A total of 530 CPR Gram-negative clinical isolates were collected from four tertiary care hospitals of Rawalpindi/Islamabad including Pakistan Institute of Medical Sciences Hospital, Federal Government Polyclinic Hospital, CDA Capital Hospital from Islamabad, and Military Hospital of Rawalpindi from the period of May 2018–September 2019. Ethical approval for this study was obtained from the Institutional review board (IRB) and Ethics committee, National University of Medical Sciences (Date: 2-9-2017/No: 06/R&D/03 NUMS). and IRB of the Shaheed Zulfikar Ali Bhutto Medical University (Date: 13-11-2017/No: F.1-1/2015/ERB/SZABMU/). Isolates were recovered from various clinical specimens which include urine, blood, pus, catheters, bronchial lavage, fluids, wound swab, and sputum.

2.2. Isolation, identification, and detection of phenotypic colistin resistance

The Clinical isolates were inoculated on MacConkey agar, Cystine lactose electrolyte deficient (CLED) agar, blood agar, and chocolate agar for 24 h at 37 °C. Their biochemical identification was done by API 10S test strip (BioMerieux Inc). Minimum inhibitory concentration (MIC) for colistin was done via broth-microdilution method as per EUCAST guidelines (<http://www.eucast.org>). Breakpoint tables for interpretation of MICs used were from Version 10.0, 2020, isolates having MIC value ≥ 4 mg/L were considered as resistant.

2.3. Antibiotic susceptibility profiling of CR isolates

All the CR isolates were subjected to antibiotic susceptibility profiling using a panel of ten antibiotics which included Piperacillin/Tazobactam, Cefazolin, Cefepime, Cefotaxime, Cefoxitin, Aztreonam, Imipenem, Amikacin, Ciprofloxacin, and Cefoperazone/sulbactam via Kirby Bauer disk diffusion method as per CLSI guidelines (M100-29th Edition <https://clsi.org/>).

2.4. Detection of *mcr* genes

Plasmid extraction was done using Gene JET Plasmid Miniprep Kit (Thermo scientific). All the phenotypic CR Gram-negative isolates were subjected to PCR for the detection of *mcr-1*, *mcr-2*, and *mcr-3* with previously described primers and cyclic conditions [19,20].

2.5. Identification of Phenotypic hvCR *K. pneumoniae*

String test was used for the identification of hv phenotypes among CR *K. pneumoniae*. The formation of (>5 mm) muco-viscous string on a loop by a bacterial colony was considered bacteriologically positive for the hv phenotype [21].

2.6. DNA extraction and Whole-genome sequencing of hvCR *K. pneumoniae*

Whole-genome of the hvCR *K. pneumoniae* isolates were extracted via the CTAB-Phenol chloroform method. Quantification of DNA via A260/A280 ratio was carried out by using a Nanodrop spectrophotometer (Titertek Berthold, Germany). Whole-genome sequencing (WGS) of two of the hvCR *K. pneumoniae* was carried out via the Illumina platform. TruSeq Nano DNA Kit was used for the library preparation while HiSeq4000 plate-form was used for whole-genome sequencing as a result 150 bp paired-end reads were generated with 100x genome coverage. *De novo* assembly of the raw reads was carried out via SPAdes v3.12.0 and unicycler while the quality was assessed via QUAST. Comprehensive Genome annotation was carried out using PATRIC (<https://www.patricbrc.org/>) and RAST tool kit. Multilocus sequence typing was carried out using <https://bigsdb.pasteur.fr> and <https://cge.cbs.dtu.dk/services/MLST/> online tools. Screening of Virulence and antibiotic determinants was done by using CGE (<https://cge.cbs.dtu.dk/services/ResFinder/>), CARD (<https://card.mcmaster.ca/analyze/rgi>) and Arg Annot (https://ifr48.timone.univ-mrs.fr/blast/arg-annot_nt.html). Plasmid Finder 2.1 (<https://cge.cbs.dtu.dk/services/PlasmidFinder/>) was used to search for the carriage of plasmid replicon. Mutations in chromosomal colistin resistance genes (like *pmrA/B*, *phoP/Q*, *mgrB*, and *eptA/B*, etc) were carried out via BLAST against the reference genome of colistin susceptible *K. pneumoniae* MGH 78578 (NC_009648.1).

3. Results

Among all the 530 CPR Gram-negative isolates, 69% of the isolates belonged to *Enterobacteriaceae* with *E. coli* (47%) and *K. pneumoniae* (46%) being the dominant members while among *Non-Enterobacteriaceae*, *A. baumannii*, and *P. aeruginosa* were 50% and 37% respectively. The Gram-negative isolates were also separated based on clinical specimens from which they were primarily isolated. Most of the isolates were from urine (42%), followed by blood (19%) and pus (17%). Among *Enterobacteriaceae*, urine was the predominant source specimen, as 91.8% (157/171) of *E. coli* and 30% (51/170) of *K. pneumoniae* were recovered from it while among *non-Enterobacteriaceae*, pus was the most common source for *A. baumannii* and *P. aeruginosa* as 37% and 29.5% were retrieved respectively.

Overall phenotypic colistin resistance was 28% in CPR Gram-negative isolates. Among *Enterobacteriaceae*, 51% (87/170) *K. pneumoniae* isolates showed phenotypic colistin resistance while only

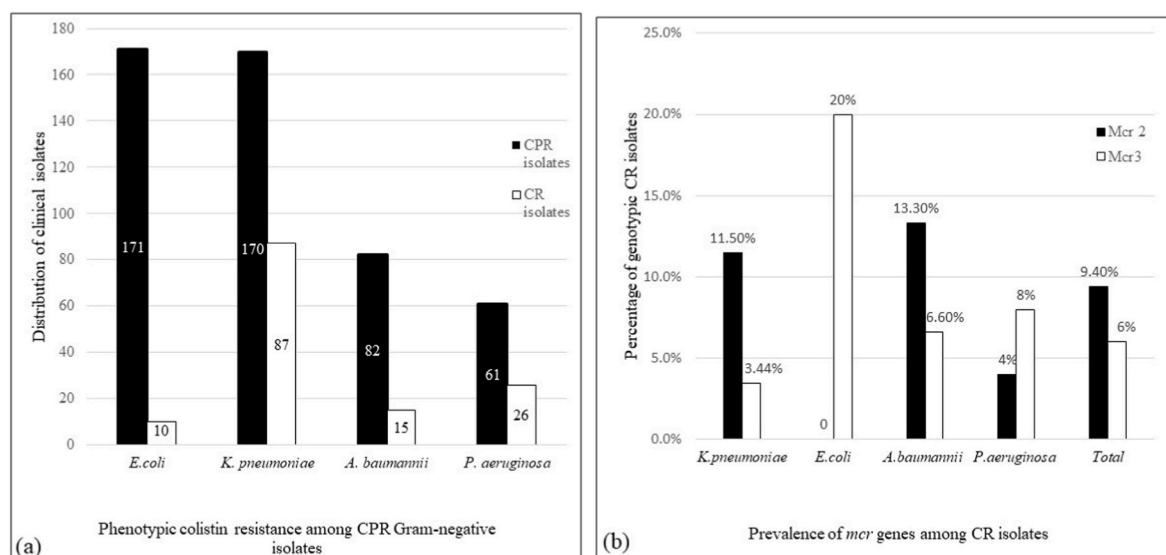


Fig. 1. (a) Distribution of carbapenem-resistant Gram-negative isolates based on their phenotypic colistin resistance, (b) Prevalence of *mcr* genes among CR Gram-negative isolates.

Table 1

Antibiotic susceptibility profiling of CR isolates among Enterobacteriaceae and non-Enterobacteriaceae.

S.No	Antibiotic tested	Enterobacteriaceae						Non-enterobacteriaceae					
		<i>K. pneumoniae</i> (%)			<i>E. coli</i> (%)			<i>A. baumannii</i> (%)			<i>P. aeruginosa</i> (%)		
		S	I	R	S	I	R	S	I	R	S	I	R
1	Piperacillin/Tazobactam	2.0	–	98.0	10.0	–	90.0	13.0	–	87.0	19.0	11.0	69.0
2	Cefazolin	2.0	–	98.0	10.0	–	90.0	13.0	–	87.0	–	–	–
3	Cefepime	3.5	–	96.5	0.0	–	100.0	13.0	–	87.0	23.0	0.0	77.0
4	Cefotaxime	1.0	1.0	98.0	0.0	–	100.0	13.0	–	87.0	–	–	–
5	Cefoxitin	2.0	–	98.0	10.0	–	90.0	13.0	–	87.0	–	–	–
6	Aztreonam	7.0	1.0	92.0	0.0	–	100.0	20.0	–	80.0	15.0	4.0	81.0
7	Imipenem	4.0	14.0	82.0	0.0	20.0	80.0	13.0	13.0	73.0	31.0	8.0	61.0
8	Amikacin	21.0	1.0	78.0	40.0	–	60.0	20.0	–	80.0	27.0	0.0	73.0
9	Ciprofloxacin	3.5	–	96.5	0.0	10.0	90.0	20.0	–	80.0	23.0	4.0	73.0
10	Cefoperazone/sulbactam	9.0	–	91.0	–	–	100.0	–	–	–	–	–	–

6% (10/171) of *E. coli* isolates showed resistance with MIC values ranging from ≥ 4 to 256 mg/L in both organisms. Among *Non-Enterobacteriaceae*, 43% (26/61) of *P. aeruginosa* isolates had MIC values ranging from ≥ 4 to 256 mg/L as compared to *A. baumannii* (15/82), which displayed MIC values range of ≥ 4 –64 mg/L (Figure 1a). Phenotypic CR isolates were detected in a higher frequency from urine followed by blood and pus specimen. 60% (6/10) of CR *E. coli* were isolated from the urine while for CR *K. pneumoniae* the most common specimen type was blood (28%) followed by urine (24%).

A high prevalence of antibiotic resistance was observed where 90% of CR isolates displayed an MDR profile. Interestingly, only 6% of *E. coli* isolates were CR while all of these were MDR. Resistance rate was high against cephalosporin, aztreonam, ciprofloxacin, and low against amikacin in CR *E. coli*. In the case of *K. pneumoniae*, about 97% of CR isolates depicted MDR profile as they showed resistance against cephalosporin, ciprofloxacin, and carbapenem, while the lowest resistance rate was observed for amikacin. The prevalence of MDR phenotype was also higher among CR *A. baumannii* (80%) and *P. aeruginosa* (73%) isolates. The highest resistance rate was observed against cephalosporin and aztreonam while lowest for the imipenem among CR *A. baumannii* and *P. aeruginosa* isolates (Table 1).

Genotypic colistin resistance was also screened in the phenotypic CR isolates. Overall carriage of *mcr*-2 and *mcr*-3 gene is presented in Fig. 1b. The prevalence of *mcr*-2 gene was found to be highest (9.4%) followed by *mcr*-3 (6%). The *mcr*-2 gene was detected in *K. pneumoniae* (10/170 of

total isolates; 10/84 of CR isolates), *A. baumannii* (2/82 of total isolate; 2/15 of CR isolates), and *P. aeruginosa* (1/61 of total isolate; 1/26 of CR isolates). The *mcr*-3 gene was detected in 3 isolates of *K. pneumoniae* (3/170 of total isolate; 3/84 of CR isolates), followed by *E. coli* (2/171 of total isolate; 2/10 of CR isolates), *P. aeruginosa* (2/61 of total isolate; 2/26 of CR isolates) and *A. baumannii* (1/82 of total isolate; 1/15 of CR isolates). While the *mcr*-1 gene was not detected in any of the phenotypic CR isolates. Co-prevalence of *mcr*-2 and 3 genes were detected in 3 *K. pneumoniae* isolates (3/170 total isolates; 3/97 CR isolates).

Among CR *K. pneumoniae* isolates, two isolates (*hv*CRKP-10718 and *hv*CRKP-4771) were selected for genomic characterization via the WGS approach as both the isolates had MIC value 8 mg/L, were negative for the presence of plasmid-mediated colistin resistance genes (*mcr*-1,2 and 3 genes), and additionally displaying a *hv*-phenotype. Isolate *hv*CRKP-10718 was isolated from pus of a 60-year-old female patient suffering from soft tissue infection and admitted to PIMS hospital Islamabad while isolate *hv*CRKP-4771 was recovered from the blood of a newborn suffering from bacteremia, admitted in Military health hospital Rawalpindi. MLST revealed that *hv*CRKP-10718 and *hv*CRKP-4771 strains belong to high-risk clones, ST-2096, and ST-147 respectively. WGS data identified various non-synonymous mutations in chromosomal genes that were involved in the modification of the outer membrane via transfer of PEA or Lipid A-Ara4N. Seven novel mutations in *eptA* (I38V; V50L; A135P), *opgE* (M53L; T486A; G236S), *arnD* (S164P) genes were also identified. Whereas mutations in *eptA* (I138V), *mcrB* (M1V), *opgE*

Table 2Phenotypic and whole genome based characteristics of hyper-mucoviscous CR-*K. pneumoniae* isolates.

S. No	Characteristics	CRKP-10718	CRKP-4771
1	Gender	Female	Male
2	Age	60	Newborn
3	Source	Pus	Blood
4	Colistin MIC (mg/L)	4	4
5	Phenotypic Resistance profile	IMP; CT; TGC; AK; ATM; CTX; FEP; CIP; CZ; FOX; TPZ; CES	IMP; CT; TGC; AK; ATM; CTX; FEP; CIP; CZ; FOX; TPZ; CES
6	Hyper-mucoviscous phenotype	+	+
7	Mobile colistin resistance genes (<i>mcr-1</i> , <i>mcr-2</i> , <i>mcr-3</i>)	Not detected	Not detected
8	Whole-genome sequencing data		
	Size (bp)	5,811,604	5,786,033
	Contig number	182	118
	GC content	56.74	56.89
	CDS	5805	5768
	tRNA	92	78
	Plasmid replicons	ColK3 ColRNAI IncFIB(K)	Col4401 ColRNAI IncFIB(K) (pCAV1099-114)
		IncFIB (pNDM-Mar) IncHI1B (pNDM-MAR)	IncL IncR
	ST	2096	147
	ybST		280
	K-locus	wzc (CD1-VR2-CD2 region of wzc) wzi (outer membrane protein of cps cluster)	K64
		K14.K64	K14.K64
9	Resistance determinants: Antibiotic resistance genes	<i>oqx</i> A; <i>oqx</i> B; <i>aac</i> (6')-Ib-cr; <i>aad</i> A2; <i>arm</i> A; <i>bla</i> _{SHV-106} ; <i>bla</i> _{OXA-232} (<i>bla</i> _{OXA-48} variant); <i>bla</i> _{TEM-1A} ; <i>bla</i> _{OXA-1} ; <i>bla</i> _{CTX-M-15} ; <i>ompK36</i> ; <i>ompK37</i> ; <i>msr</i> (E); <i>cat</i> B3; <i>tet</i> (D); <i>mph</i> (E); <i>msr</i> (E); <i>fos</i> A; <i>sul</i> 1; <i>dfr</i> A1; <i>dfr</i> A12; <i>dfr</i> A14 <i>ompK36</i> : <i>g231_None232insG</i> ; <i>ompK36</i> :p.H235 N; <i>ompK36</i> : <i>y306_None307insGY</i> ?; <i>ompK36</i> : <i>g305_None306insR</i> ; <i>ompK36</i> : <i>l230_None231insAL</i> ; <i>ompK36</i> :p.D310T; <i>ompK36</i> :p.Q228_None682del; <i>ompK36</i> :p.Q227_None679del; <i>ompK36</i> : <i>g133_None134insGGD</i> ; <i>ompK36</i> :p.A190_None568del; <i>ompK36</i> :p.D308T; <i>ompK36</i> :p.V202L; <i>ompK36</i> :p.G307A; <i>ompK36</i> : <i>l229_None230insP</i> ; <i>ompK36</i> :p.I311S; <i>ompK36</i> :	<i>qnrS1</i> ; <i>oqx</i> A; <i>rmtF</i> ; <i>aac</i> (6')-Ib-cr; <i>aac</i> (6')-Ib- <i>Hangzhou</i> ; <i>aph</i> (3')-Ib; <i>aph</i> (6)-Ib; <i>bla</i> _{SHV-11} ; <i>bla</i> _{OXA-48} ; <i>bla</i> _{TEM-1} ; <i>bla</i> _{CTX-M-15} ; <i>bla</i> _{SHV-67} ; <i>tet</i> (A); <i>arr-2</i> ; <i>fos</i> A; <i>sul</i> 1; <i>dfr</i> A1
	Point mutations Conferring resistance to antimicrobials		<i>ompK37</i> (p.I128 M); <i>ompK37</i> (p.I70 M); <i>acrR</i> (p.L195V); <i>acrR</i> (p.G164A); <i>acrR</i> (p.R173G); <i>acrR</i> (p.F172S); <i>acrR</i> (p.F197I); <i>acrR</i> (p.K201 M); <i>acrR</i> (p.P161R)

Table 2 (continued)

S. No	Characteristics	CRKP-10718	CRKP-4771
		<i>k192_None193insK</i> ; <i>ompK36</i> :p.A280V; <i>ompK36</i> :p.L312*; <i>ompK36</i> :p.Q309R; <i>ompK36</i> : <i>l230_None231insAL</i> ; <i>ompK36</i> :p.G232_None696del; <i>ramR</i> :p.K194*; <i>ramR</i> :p.K9I; <i>acrR</i> (p.G164A); <i>acrR</i> (p.P161R); <i>acrR</i> (p.F172S); <i>acrR</i> (p.F197I); <i>acrR</i> (p.L195V); <i>acrR</i> (p.R173G); <i>acrR</i> (p.K201 M)	
	Metal resistant genes	<i>merACDEPRT</i> ; <i>silA</i> ; <i>terB</i> ; <i>terD</i> ; <i>terE</i> ; <i>terW</i> ; <i>terX</i> ; <i>terZ</i>	<i>pcoABCDERS</i> ; <i>silABCEFGPRS</i>
10	Virulence determinants:		
	Iron Acquisition system	<i>Yersinia</i> <i>irp1</i> ; <i>irp2</i> ; <i>YbtAESTPQUX</i> ; <i>fyuA</i>	<i>irp1</i> ; <i>irp2</i> ; <i>ybtAESTPQUX</i> ; <i>fyuA</i>
	Enterobactin	<i>entABES</i> ; <i>fepBCDG</i>	<i>entAB</i> ; <i>fepB C D G</i>
	Aerobactin system	<i>iucABCDT</i> ; <i>iroN</i> <i>KfuABC</i> Present	Absent Absent
	Adhesions	<i>yagY/ecpB</i> ; <i>yagW/ecpD</i> ; <i>yagZ</i> ; <i>yagZ/ecpA</i> ; <i>ykgk/ecpR</i> ; <i>yagV/ecpE</i> ; <i>yagX/ecpC</i>	<i>yagY/ecpB</i> ; <i>yagW/ecpD</i> ; <i>yagZ</i> ; <i>yagZ/ecpA</i> ; <i>ykgk/ecpR</i> ; <i>yagV/ecpE</i> ; <i>yagX/ecpC</i>
	Adherence and Invasion	<i>fimABCEH</i>	<i>fimABCDEH</i>
	<i>RmpA/A2</i> Capsule regulatory genes	Present with Frame shift mutation	Absent
	Fimbrial genes	<i>mrkABCDHFHJ</i>	<i>mrkABCDHFHJ</i>
	Biofilm	<i>sugE</i> ; <i>aroA</i>	<i>sugE</i>

(M53L; T486A), and *arnT* (R372K) were detected in both the isolates. The impact of mutations was analyzed via PROVEAN (http://provean.jcvi.org/seq_submit.php) and PREDICT SNP (<https://loschmidt.chemi.muni.cz/predictsnp1/>). Mutation in S257L (*eptA*) and F170I (*pagP*) in *hvCRKP-10718* and G236S (*opgE*) in *hvCRKP-4771* were predicted to be deleterious. Various single nucleotide changes were also identified along with the non-synonymous mutations, all the variations identified in this study are presented in Table 3. Apart from the acquisition of mutations, WGS data identified the insertion of 6.6 kb foreign DNA segment via mobile elements mainly transposases (IS4 and IS2) and prophage P4 integrase in *hvCRKP-4771* isolate which carried two putative PEA transferase *yjgX* genes. (Fig. II a). Both the putative PEA transferases (located at position 2980–3411; 3535–3777) showed 100% sequence homology with *E. coli* PEA transferases, additionally sequence alignment of 6.6 Kb segment from Blast database also showed 100% sequence similarity with *E. coli* and *Shigella* sp indicating toward the possible source for the acquisition of this segment (Fig. II b).

WGS data reveals that both the isolates possessed multiple resistant determinants depicting an AMR profile. Genes encoding β -lactamases like *bla*_{OXA-1}, *bla*_{OXA-232}, *bla*_{TEM-1A}, *bla*_{CTX-M-15}, *bla*_{SHV-106} were identified in *hvCRKP-10718* while, *bla*_{OXA-48}, *bla*_{TEM-1}, *bla*_{CTX-M-15}, *bla*_{SHV-11}, *bla*_{SHV-67} genes were identified in *hvCRKP-4771* isolate. Carbapenem resistance among *hvCRKP-10718* isolate was associated with *bla*_{OXA-232} (a variant of *bla*_{OXA-48}) along with various mutations in *OmpK36* and *OmpK37* genes while in the case of Isolate *hvCRKP-4771* resistance against carbapenem was due to *bla*_{OXA-48} and mutations in *OmpK37* gene (Table 2). Additionally, Both the isolates were resistant to tetracycline and fosfomycin by harboring *tet* and *fosA* genes, respectively. Metal resistance

Table 3

Chromosomal variations identified in genes that might be responsible to confer resistance against colistin among Hyper-mucoviscous CR-*K. pneumoniae* isolates via Whole-genome sequencing.

Colistin resistant determinants		Single nucleotide Variation		Amino acid variation		
		CRKP_10718	CRKP-4771	CRKP-10718	CRKP-4771	
Lipid A modification via PEA transfer	<i>phoP</i>	207 T > C	207 T > C	None	None	
		363 T > C	363 T > C			
		633 C > T	465 T > C			
	<i>phoQ</i>	C81G	81 C > G	None	None	
		C309A	1194 C > T			
		C1023T				
		T1185C				
	<i>pmrA</i>	Wild type		Wild type		
		<i>pmrB</i>	264 A > T			663C > G
	<i>eptA</i>		306 T > C	846 A > G		R256G
			1002 A > C			
			None	153 C > T	I138V	C27F
				273 C > T	S257L ^b	V50L^a
	<i>eptB</i>			1030 C > T		A135P^a
			1347T > C		I138V	
			1410 G > A			
		507 G > A	507G > A	None	None	
<i>mgrB</i>		747 C > T	642A > T			
			651C > T			
			933C > T			
		None	None	M1V	M1V	
<i>opgE</i>		None	162 G > C	M53L^a	M53L	
			585C > T	T486A	G236S^b	
			1491 A > G		T486A^a	
			G			
<i>pagP</i>		138 C > T	297 C > G	F170I ^b	None	
		183 G > A				
	Lipid A modification via A-Ara4N transfer	Undecaprenyl-phosphate 4-deoxy-4-formamido-1-arabinose transferase	504 C > T	54 C > T	None	None
			642 T > C	75 A > G		
624 T > C			162 C > G			
783 C > T			378 C > T			
789 A > G			624 T > C			
Undecaprenyl phosphate-alpha-4-amino-4-deoxy-1-arabinose arabinosyl transferase (<i>arnT</i>)			774 C > T			
			783 C > T			
			789 A > G			
			None	M114L	H156Q	
			636 G > A	V117I	R372K	
4-deoxy-4-formamido-1-arabinose-phosphoundecaprenol deformylase (<i>arnD</i>)	682		R372K			
	C > T					
	843 T > C					
	951 C > T					
Undecaprenyl phosphate-aminoarabinose flippase subunit (<i>arnE</i>)	370 A > C	735 T > C	S164P ^a	None		
	None	None	None	None		
	None	None	None	None		

The bold alphabet represent amino acid as I(Isoleucine at position 138 in eptA gene replaced valine) while the exponent value "a" represent novel non-synonymous mutation.

^a Indicates novel non-synonymous mutations.

^b Indicates the deleterious impact of the mutation.

genes were also identified in hvCRKP-10718 like mercury and tellurium resistant genes while copper and silver resistant determinants were identified in hvCRKP-4771. Apart from these, other multiple resistant determinants were also detected among both the isolates conferring resistance to most of the antibiotics. Furthermore, both the isolates depicted a hypervirulent profile as several virulence determinants were also detected responsible for a hypervirulent trait (Table 2).

4. Discussion

Extensive use of antimicrobials for the treatment of MDR Gram-negative infection restricted treatment choices, hence leaving colistin as the last remaining option [3]. However, resistance toward colistin 'last resort antibiotic' was reported in CPR Gram-negative pathogens causing difficulty in treating infections worldwide [22]. From Pakistan, sporadic studies from multiple hospitals identified an upsurge of colistin resistance among Gram-negative bacteria. No data is available on the

overall prevalence of colistin resistance among CPR Gram-negative bacteria which might be attributed to the lack of a centralized surveillance system in our country. In this study, more than one-fourth (28%) of clinical isolates showed resistance to colistin which is approximately double as reported 11.9% in 2020 among Gram-negative clinical isolates [23]. The prevalence of colistin resistance was higher in *K. pneumoniae* (51%) as compared to *E. coli* (6%) isolates. These results were in line with previous literature in which CR rates differ among *Enterobacteriaceae* as *K. pneumoniae* showed higher resistance as compared to *E. coli* worldwide [2]. A study from Pakistan also reported concurrent findings in which the phenotypic colistin resistance was reported 50% in *K. pneumoniae* isolates and only 7% among *E. coli* [24]. In the present study, *P. aeruginosa* showed higher phenotypic colistin resistance (43%) as compared to *A. baumannii* (18%) these findings were concurrent with the recent studies, that reported a higher prevalence of CR (21.3%) in Egypt [25] and (11.9%) in Pakistan [16] among *P. aeruginosa*. while *A. baumannii* prevalence was 17.5% in Lebanon, 15% in China [26] and

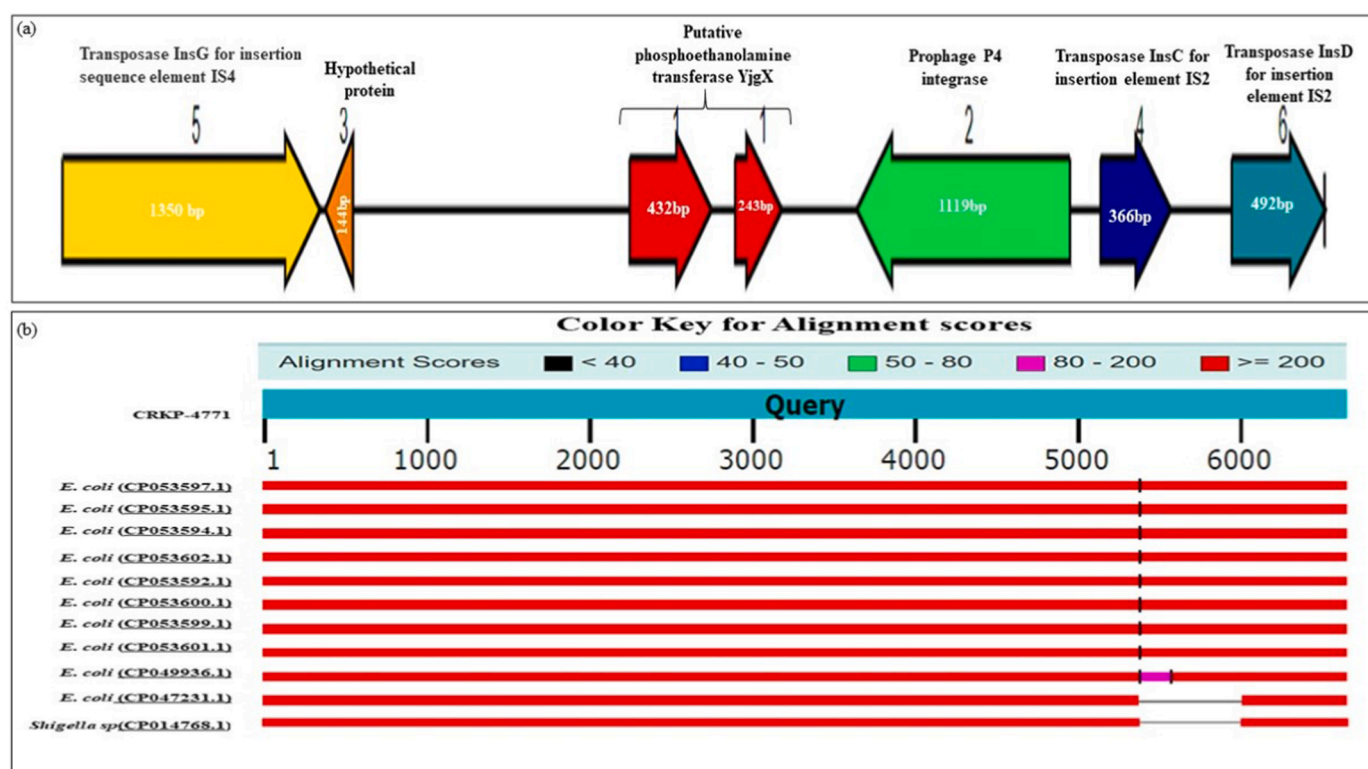


Fig. 2. (a) represented the acquisition of 6.6 kb mobile genetic element equipped with putative PEA transferases (YjgX) acquired by hvCRKP-4771 isolate, (b) Depicts the alignment scores of the acquired mobile genetic element of hvCRKP-4771 in comparison with different Gram-negative organisms by using Blast database. The blue bar indicates the query sequence while the red bar indicates the significant matches of respective Gram-negative organisms reported in the database.

9.6% in Pakistan [16].

Previous studies identified a drastic increase in antibiotic resistance and the development of MDR in Pakistan [10]. Similarly, a high prevalence of MDR profile was also observed in our study as 90% of Gram-negatives were identified displaying MDR phenotype among CR isolates. This escalating trend in MDR prevalence has also recently been reported in many studies from Pakistan in *E. coli* [13], *K. pneumoniae* [27], *A. baumannii* [16,26], and *P. aeruginosa* [12,16,25]. This high prevalence of MDR Gram-negative pathogens in health care settings may be attributed to the inappropriate and over-the-counter use of broad-spectrum antibiotics in healthcare settings [28]. In Pakistan, MDR *K. pneumoniae* isolates with Classical ST-2096 [29] and ST-147 [30] have been reported however both the isolates from the current study differed from the classical ST-2096 and ST-147 because the additional acquisition of antimicrobial and virulent determinants representing a convergent profile concurrent with the recent studies from India [31] and Saudi Arabia [32]. Southeast Asia is now considered a hotspot region as several recent studies reported different high-risk *K. pneumoniae* convergent clones from healthcare settings reflecting the possible clonal spread of this emerging superbug [18,31].

MDR pathogens confer resistance to colistin mainly via modification of the outer membrane by addition PEA or Lipid A-Ara4N either through chromosomal alteration of the lipid-A modification genes or via the acquisition of *mcr* genes [33]. Based on PCR analysis, *mcr-2* gene was detected in 9.4% ($n = 13$) CR isolates while *mcr-3* carriage was observed in 6% ($n = 8$) isolates. In the current study, carriage of *mcr-2* gene was observed in *K. pneumoniae*, *A. baumannii*, and *P. aeruginosa* isolates which differ from the previously reported studies that reported *mcr-2* gene only in *E. coli* and *Salmonella* species exclusively in European countries [34,35]. Most recently, Imtiaz et al. [18], detected *mcr-2* gene carriage in *K. pneumoniae* clinical isolates from Pakistan. The carriage of *mcr-3* gene in *E. coli* and *K. pneumoniae* was reported in many studies [36,37]. Moreover, *mcr-3* carriage in *P. aeruginosa* was recently reported

from ocular samples in India [38] while not a single study reported the presence of *mcr-3* gene in *A. baumannii* isolates [39]. The dissemination of *mcr-2* and *mcr-3* gene among these pathogens might be associated with excessive colistin use in our local poultry, agriculture, and even in clinical settings. As *mcr* genes are located on the mobile genetic element hence are highly prevalent among animal and environmental samples [40]. Horizontal transfer of these genes among Gram-negative pathogens from environmental/animal samples might have paved the way to human transmission via food chain or fecal to oral route [2,6]. Although, *mcr* gene prevalence was low in our study but there was a high prevalence of phenotypic CR (28.5%) pointing towards other possible mechanisms that might be responsible for colistin resistance in these pathogens [2].

The acquisition of mutations in chromosomal determinants involved in modification of lipid-A (*phoP/Q*, *pmrA/B*, *mgrB*, *pagP*, *arn* operon, *eptA*, and *eptB*, etc), have been previously reported to play important role in the development of colistin resistance even in MDR Gram-negative pathogens that lack plasmid-encoded *mcr* genes [5,41,42]. Likewise, in our study two of CR *K. pneumoniae* isolates characterized via WGS which were phenotypically resistant to colistin with MIC value-8mg/L and were negative for *mcr* genes have acquired various non-synonymous chromosomal mutations in genes involved in modification of the outer membrane which includes *eptA* (I138V), *mgrB* (M1V), *opgE* (M53L), and *arnT* (R372I). Mutations in *pmrB* (T246A; R256G), *arnT* (M114L; V117I; R372K; H156Q) and *pagP* gene (F170I) were concurrent with previous studies from India [5,41]. Furthermore, mutations in *mgrB* (M1V) [5,43] and *eptA* (C27F; S257L) [43] were also reported previously. Apart from all the non-synonymous mutations, 7 variations in Lipid A modification genes [*eptA* (I38V; V50L; A135P), *opgE* (M53L; T486A; G236S), *arnD* (S164P)] have not been previously reported and were novel to this study. Apart from the presence of chromosomal encoded PEA transferases (EptA/B; OpgE), insertion of putative PEA transferases YjgX in hvCRKP-4771 was identified for the first time in CR-*K. pneumoniae*. The

yjgX gene has been initially identified in *E. coli* K-12 strain [44]. A recent study by Saile et al., (2018), reported the integration of foreign DNA segments via mobile genetic elements mainly transposases and phage integrases that carried putative PEA transferase gene (*yjgX*) in a food-borne Shiga toxin-producing *E. coli*. The integration of *yjgX* gene was found to be associated with pathogenicity in Shiga toxin-producing *E. coli* isolates [45]. However, the exact role of *yjgX* gene encoding putative PEA transferase needs to be further evaluated in colistin resistance *mcr*-negative isolates. In contrast to previous findings, this is the first study to report the integration of the putative PEA transferase (*yjgX*) gene via mobile genetic elements, a possible mechanism of acquisition and inter specie spread of this putative transferase in *hVCR-K. pneumoniae* isolate. The presence of novel mutations and acquisition of putative PEA transferases by colistin-resistant isolates depicts the association of these genetic determinants as a possible chromosomal mechanism of colistin resistance and enhanced virulence among these evolving superbugs.

One of the limitation of the current study was PCR based screening of only three mobile colistin resistant genes (*mcr*-1, 2 and 3), although other subtypes of *mcr* genes might also be involved in colistin resistance which needs to be determined in these isolates. Moreover, due to financial constraints, this study was restricted to WGS based characterization of only 2 CR-K. *pneumoniae* isolates. Further large-scale studies are needed to decipher the role of chromosomal mutations in colistin resistance among Gram-negative superbugs in Pakistan.

5. Conclusion

This study highlighted high phenotypic colistin resistance, but comparatively low plasmid-mediated gene carriage especially in CR K. *pneumoniae*, indicating other possible mechanisms of colistin resistance. The current study is the first to report emergence and whole genome-based characterization of high-risk convergent CR K. *pneumoniae* clones from Pakistan harboring several chromosomal mutations and genetic determinants associated with colistin resistance. As per our knowledge, mutations in genes that mediate Lipid-A modification (like *pmrA/B*, *phoP/Q*, *mgrB*, and *eptA/B*, etc) and possible acquisition of *yjgX* gene have not been previously studied in Pakistan. Collectively, these findings highlighted the clinical importance of the diverse mechanism of resistance against colistin. The contribution of all these factors further needs large-scale comprehensive genome-based studies to evaluate other possible mechanisms of resistance to this last resort drug.

ICMJE statement

HMM, SA, AM and RF conceived presented idea. MU and RU performed experiments and analyzed MIC data. Whole genome Sequencing was performed and analyzed by MU and RF. RF supervised this work. Manuscript was written by MU and reviewed by all authors. All authors contributed to the final manuscript.

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Data availability

This Whole Genome Shotgun projects of CRKP-10718 and CRKP-4771 have been deposited and now been made public at DDBJ/ENA/GenBank under the accession JAFIDU000000000 and JAHARGO000000000, respectively. The versions described in this paper are version JAFIDU010000000 and JAHARGO010000000, respectively.

Declaration of competing interest

All the authors declare no conflict of interest.

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References

- [1] Petrosillo N. Infections: the emergency of the new millennium. Nucl. Med. Infect. Dis. 2020;1–8. https://doi.org/10.1007/978-3-030-25494-0_1. Springer International Publishing.
- [2] Li Z, Cao Y, Yi L, Liu JH, Yang Q. Emergent polymyxin resistance: end of an Era? Open Forum Infect Dis 2019;6. <https://doi.org/10.1093/ofid/ofz368>.
- [3] Laxminarayan R, Van Boeckel T, Frost I, Kariuki S, Khan EA, Limmathurotsakul D, et al. The lancet infectious diseases commission on antimicrobial resistance: 6 years later. Lancet Infect Dis 2020;20:e51–60. [https://doi.org/10.1016/S1473-3099\(20\)30003-7](https://doi.org/10.1016/S1473-3099(20)30003-7).
- [4] Sundaramoorthy NS, Mohan HM, Subramaniam S, Raman T, Selva Ganesan S, Sivasubramanian A, et al. Ursolic acid inhibits colistin efflux and curtails colistin resistant Enterobacteriaceae. AMB Express 2019;9. <https://doi.org/10.1186/s13568-019-0750-4>.
- [5] Mathur P, Veeraraghavan B, Devanga Ragupathi NK, Inbanathan FY, Khurana S, Bhardwaj N, et al. Multiple mutations in lipid-A modification pathway & novel fosA variants in colistin-resistant Klebsiella pneumoniae. Futur Sci OA 2018;4. <https://doi.org/10.4155/fsoa-2018-0011>.
- [6] Liu YY, Wang Y, Walsh TR, Yi LX, Zhang R, Spencer J, et al. Emergence of plasmid-mediated colistin resistance mechanism MCR-1 in animals and human beings in China: a microbiological and molecular biological study. Lancet Infect Dis 2016;16:161–8. [https://doi.org/10.1016/S1473-3099\(15\)00424-7](https://doi.org/10.1016/S1473-3099(15)00424-7).
- [7] Yang YQ, Li YX, Lei CW, Zhang AY, Wang HN. Novel plasmid-mediated colistin resistance gene mcr-7.1 in Klebsiella pneumoniae. J Antimicrob Chemother 2018;73:1791–5. <https://doi.org/10.1093/jac/dky111>.
- [8] Kluytmans J. Plasmid-encoded colistin resistance: Mcr-one, two, three and counting. Euro Surveill 2017;22. <https://doi.org/10.2807/1560-7917.ES.2017.22.31.30588>.
- [9] Sabir R, Faraz Danish Alvi S, Fawwad A. Antimicrobial susceptibility pattern of aerobic microbial isolates in a clinical laboratory in Karachi - Pakistan. Pakistan J Med Sci 2012;29. <https://doi.org/10.12669/pjms.293.3187>.
- [10] Sohail Afzal M. Emergence of antibiotic resistance in Pakistan; A clear problem for future. J Vaccines Vaccin 2017;8. <https://doi.org/10.4172/2157-7560.1000375>.
- [12] Farooq L, Memon Z, Ismail MO, Sadiq S. Frequency and antibiogram of multi-drug resistant pseudomonas aeruginosa in a tertiary care hospital of Pakistan. Pakistan J Med Sci 2019;35:1622–6. <https://doi.org/10.12669/pjms.35.6.930>.
- [13] Nosheen S, Bukhari NI, Ejaz H, Abbas N. Antibiogram and recent incidence of multi-drug resistant carbapenemase producing Escherichia coli isolated from paediatric patients. Pakistan J Med Sci 2020;36:246–50. <https://doi.org/10.12669/pjms.36.2.928>.
- [16] Hameed F, Khan MA, Muhammad H, Sarwar T, Bilal H, Rehman TU. Plasmid-mediated mcr-1 gene in Acinetobacter baumannii and Pseudomonas aeruginosa: first report from Pakistan. Rev Soc Bras Med Trop 2019;52:1–6. <https://doi.org/10.1590/0037-8682-0237-2019>.
- [17] Bilal H, Hameed F, Khan MA, Khan S, Yang X, Rehman TU. Detection of mcr-1 gene in extended-spectrum β -lactamase-producing Klebsiella pneumoniae from human urine samples in Pakistan. Jundishapur J Microbiol 2020;13:1–6. <https://doi.org/10.5812/jjm.96646>.
- [18] Imtiaz W, Syed Z, Rafeeq Z, Andrews SC, Dasti JI. Analysis of antibiotic resistance and virulence traits (Genetic and phenotypic) in klebsiella pneumoniae clinical isolates from Pakistan: identification of significant levels of carbapenem and colistin resistance. Infect Drug Resist 2021;14:227–36. <https://doi.org/10.2147/IDR.S293290>.
- [19] Yin W, Li H, Shen Y, Liu Z, Wang S, Shen Z, et al. Novel Plasmid-Mediated Colistin Resistance Gene mcr-3 in Escherichia coli. MBio 2017;8:e00543. <https://doi.org/10.1128/mBio.00543-17>.
- [20] Cavaco L, Mordhorst H, Hendriksen R. Laboratory protocol: PCR for plasmid-mediated colistin resistance genes. DTU Food Natl Food Inst 2016;2:1–15.
- [21] Shon AS, Bajwa RPS, Russo TA. Hypervirulent (hypermucoviscous) Klebsiella Pneumoniae: a new and dangerous breed. Virulence 2013;4:107–18. <https://doi.org/10.4161/viru.22718>.
- [22] Olaitan AO, Morand S, Rolain JM. Emergence of colistin-resistant bacteria in humans without colistin usage: a new worry and cause for vigilance. Int J Antimicrob Agents 2016;47:1–3. <https://doi.org/10.1016/j.ijantimicag.2015.11.009>.
- [23] Javed H, Saleem S, Zafar A, Ghafoor A, Shahzad A, Bin Ejaz H, et al. Emergence of plasmid-mediated mcr genes from Gram-negative bacteria at the human-animal interface. Gut Pathog 2020;12:1–9. <https://doi.org/10.1186/s13099-020-00392-3>.
- [24] Qamar S, Shaheen N, Shakoor S, Farooqi J, Jabeen K, Hasan R. Frequency of colistin and fosfomycin resistance in carbapenem-resistant Enterobacteriaceae

- from a tertiary care hospital in Karachi. *Infect Drug Resist* 2017;10:231–6. <https://doi.org/10.2147/IDR.S136777>.
- [25] Abd El-Baky RM, Masoud SM, Mohamed DS, Waly NGFM, Shafik EA, Mohareb DA, et al. Prevalence and some possible mechanisms of colistin resistance among multidrug-resistant and extensively drug-resistant *Pseudomonas aeruginosa*. *Infect Drug Resist* 2020;13:323–32. <https://doi.org/10.2147/IDR.S238811>.
- [26] Pormohammad A, Mehdinejadani K, Gholizadeh P, Mohtavinejad N, Dadashi M, Karimaei S, et al. Global prevalence of colistin resistance in clinical isolates of *Acinetobacter baumannii*: a systematic review and meta-analysis. *Microb Pathog* 2020;139. <https://doi.org/10.1016/j.micpath.2019.103887>.
- [27] Nahid F. Molecular epidemiology and genetic characterization of multi-drug resistant *Klebsiella pneumoniae* n.d..
- [28] Malik F, Figueras A. Continuous rise in cephalosporin and fluoroquinolone consumption in Pakistan: a 5 year analysis (2014–18). *JAC-Antimicrobial Resist* 2019;1:1–4. <https://doi.org/10.1093/jacamr/dlz063>.
- [29] Rahmat Ullah S, Majid M, Andleeb S. Draft genome sequence of an extensively drug-resistant neonatal *Klebsiella pneumoniae* isolate harbouring multiple plasmids contributing to antibiotic resistance. *J Glob Antimicrob Resist* 2020;23:100–1. <https://doi.org/10.1016/j.jgar.2020.08.008>.
- [30] Gondal AJ, Saleem S, Jahan S, Choudhry N, Yasmin N. Novel carbapenem-resistant *Klebsiella pneumoniae* ST147 coharboring blaNDM-1, blaOXA-48 and extended-spectrum β -lactamases from Pakistan. *Infect Drug Resist* 2020;13:2105–15. <https://doi.org/10.2147/IDR.S251532>.
- [31] Shankar C, Vasudevan K, Jacob JJ, Baker S, Isaac BJ, Neeravi AR, et al. Mosaic antimicrobial resistance/virulence plasmid in hypervirulent ST2096 *Klebsiella pneumoniae* in India: the rise of a new superbug? *BioRxiv* 2020. <https://doi.org/10.1101/2020.12.11.422261>.
- [32] Hala S, Antony CP, Alshehri M, Alsaedi A, Al Thaqafi OA, Al-Ahmadi G, et al. An emerging clone (ST2096) of *Klebsiella pneumoniae* clonal complex 14 with enhanced virulence causes an outbreak in Saudi Arabia. *J Infect Public Health* 2020;13:363–4.
- [33] Olaitan AO, Morand S, Rolain JM. Mechanisms of polymyxin resistance: acquired and intrinsic resistance in bacteria. *Front Microbiol* 2014;5. <https://doi.org/10.3389/fmicb.2014.00643>.
- [34] Xavier BB, Lammens C, Ruhel R, Malhotra-Kumar S, Butaye P, Goossens H, et al. Identification of a novel plasmid-mediated colistin resistance gene, mcr-2, in *Escherichia coli*, Belgium, June 2016. *Euro Surveill* 2016;21. <https://doi.org/10.2807/1560-7917.ES.2016.21.27.30280>.
- [35] Garcia-Graells C, De Keersmaecker SCJ, Vanneste K, Pochet B, Vermeersch K, Roossens N, et al. Detection of plasmid-mediated colistin resistance, mcr-1 and mcr-2 genes, in *Salmonella* spp. isolated from food at retail in Belgium from 2012 to 2015. *Foodb Pathog Dis* 2018;15:114–7. <https://doi.org/10.1089/fpd.2017.2329>.
- [36] Roer L, Hansen F, Stegger M, Sönksen UW, Hasman H, Hammerum AM. Novel mcr-3 variant, encoding mobile colistin resistance, in an ST131 *Escherichia coli* isolate from bloodstream infection, Denmark, 2014. *Euro Surveill* 2017;22. <https://doi.org/10.2807/1560-7917.ES.2017.22.31.30584>.
- [37] Haddadj L, Baron SA, Olaitan AO, Morand S, Rolain JM. Co-occurrence of variants of mcr-3 and mcr-8 genes in a *Klebsiella pneumoniae* isolate from Laos. *Front Microbiol* 2019;10. <https://doi.org/10.3389/fmicb.2019.02720>.
- [38] Mitra S, Basu S, Rath S, Sahu SK. Colistin resistance in Gram-negative ocular infections: prevalence, clinical outcome and antibiotic susceptibility patterns. *Int Ophthalmol* 2020;1–11. <https://doi.org/10.1007/s10792-020-01298-4>.
- [39] Lima WG, Alves MC, Cruz WS, Paiva MC. Chromosomally encoded and plasmid-mediated polymyxins resistance in *Acinetobacter baumannii*: a huge public health threat. *Eur J Clin Microbiol Infect Dis* 2018;37:1009–19. <https://doi.org/10.1007/s10096-018-3223-9>.
- [40] Mohsin M, Van Boeckel TP, Saleemi MK, Umair M, Naseem MN, He C, et al. Excessive use of medically important antimicrobials in food animals in Pakistan: a five-year surveillance survey. *Glob Health Action* 2019;12. <https://doi.org/10.1080/16549716.2019.1697541>.
- [41] Pragasa AK, Shankar C, Veeraghavan B, Biswas I, Nabarro LEB, Inbanathan FY, et al. Molecular mechanisms of colistin resistance in *Klebsiella pneumoniae* causing bacteremia from India-A first report. *Front Microbiol* 2017;7. <https://doi.org/10.3389/fmicb.2016.02135>.
- [42] Zhang H, Srinivas S, Xu Y, Wei W, Feng Y. Genetic and biochemical mechanisms for bacterial lipid A modifiers associated with polymyxin resistance. *Trends Biochem Sci* 2019;44:973–88. <https://doi.org/10.1016/j.tibs.2019.06.002>.
- [43] Boszczowski I, Salomão MC, Moura ML, Freire MP, Guimarães T, Cury AP, et al. Multidrug-resistant *Klebsiella pneumoniae*: genetic diversity, mechanisms of resistance to polymyxins and clinical outcomes in a tertiary teaching hospital in Brazil. *Rev Inst Med Trop Sao Paulo* 2019;61. <https://doi.org/10.1590/s1678-9946201961029>.
- [44] Blattner FR, Plunkett G, Bloch CA, Burland V, Riley M, et al. The complete genome sequence of *Escherichia coli* K-12. *Science* 1997;277:1453–62. <https://doi.org/10.1126/science.277.5331.1453>.
- [45] Saile N, Schuh E, Semmler T, Eichhorn I, Wieler LH, Bauwens A, et al. Determination of virulence and fitness genes associated with the pheU, pheV and selC integration sites of LEE-negative food-borne Shiga toxin-producing *Escherichia coli* strains. *Gut Pathog* 2018;10:43. <https://doi.org/10.1186/s13099-018-0271-8>.